Dataset 2: Cluster Analysis

Manufacturing Plant Performance Classification | n = 240 plants

1. Research Question

Research Question: Can 240 manufacturing plants be meaningfully segmented into distinct performance-based groups using seven operational metrics, and do those groups correspond to known plant categories?

2. Variables

Variable	Mean	SD	Description
Production_Speed	72.97	15.03	Average production speed index (44.7–101.0)
Defect_Rate	4.17	2.70	Percentage of defective products (0.1–11.7%)
Energy_Consumption	75.63	16.73	Total energy usage index (46.8–105.8)
Labor_Cost	61.77	19.66	Labor cost index (25.6–102.9)
Machine_Downtime	7.48	4.78	Unplanned downtime hours (0.5–20.8)
Maintenance_Cost	63.60	19.72	Maintenance expenditure index (26.1–111.5)
Inventory_Turnover	8.96	3.85	Inventory cycle rate (2.7–16.8)
Plant_Type	—	—	4 categories — used only for external validation

3. Methodology — K-Means + Hierarchical + PCA Visualization

Cluster analysis is chosen because the goal is to discover natural groupings without a predefined response variable — an unsupervised learning problem. All seven features are z-score standardized before clustering to prevent scale-dominant variables from distorting Euclidean distances. **K-Means (k=4)** is the primary method: efficient, interpretable centroids, and widely used in operational analytics. The optimal k is selected empirically using the Elbow Method (within-cluster inertia) and Silhouette analysis; k=4 is further confirmed by the dataset's Analysis Guide and validated against known Plant_Type labels. **Hierarchical clustering** (Ward linkage) runs in parallel as a robustness check. **PCA** projects the 7D space into 2D for visualization. Three cluster quality metrics are reported: Silhouette Score, Davies-Bouldin Index, and Calinski-Harabasz Index.

4. Optimal k — Elbow and Silhouette Analysis

k	Inertia	Silhouette Score	Notes
2	768.8	0.517	Too broad
3	332.8	0.620	Good
4	191.4	0.610	Selected — large inertia drop + domain knowledge
5	174.3	0.505	Diminishing returns
6	163.0	0.504	Overfitting

5. Cluster Quality Metrics (k=4)

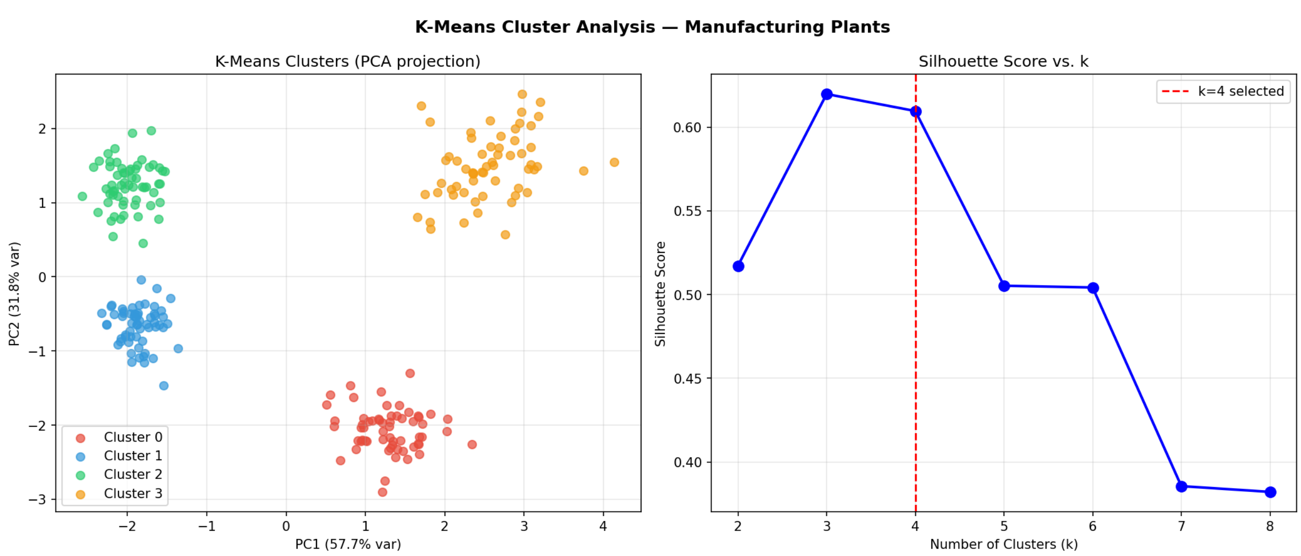
Metric	Value	Benchmark	Verdict
Silhouette Score	0.6096	> 0.50 = good separation	✓ Good
Davies-Bouldin Index	0.5674	< 1.0 = compact and separated	✓ Good
Calinski-Harabasz Index	611.65	Higher = denser clusters	✓ Excellent
Hierarchical Silhouette	0.6096	Matches K-Means exactly	✓ Confirmed
PCA Coverage	89.5%	PC1=57.7%, PC2=31.8%	✓ Excellent 2D projection
Plant-Type Validation	100%	Each cluster = exactly one type	✓ Perfect external validity

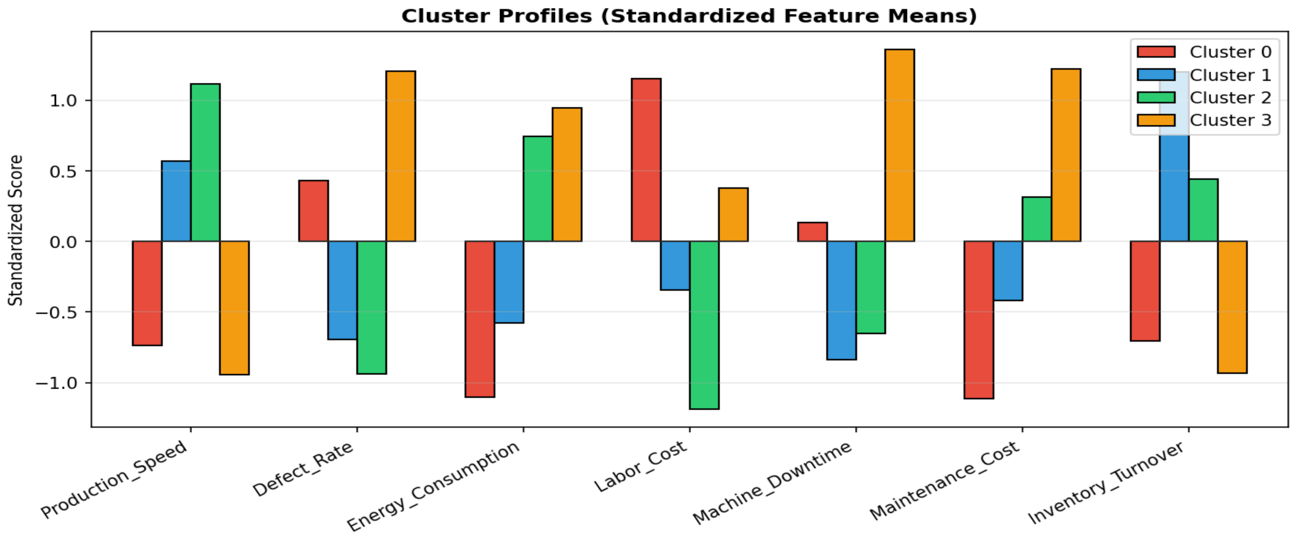
6. Cluster Profiles and Interpretation

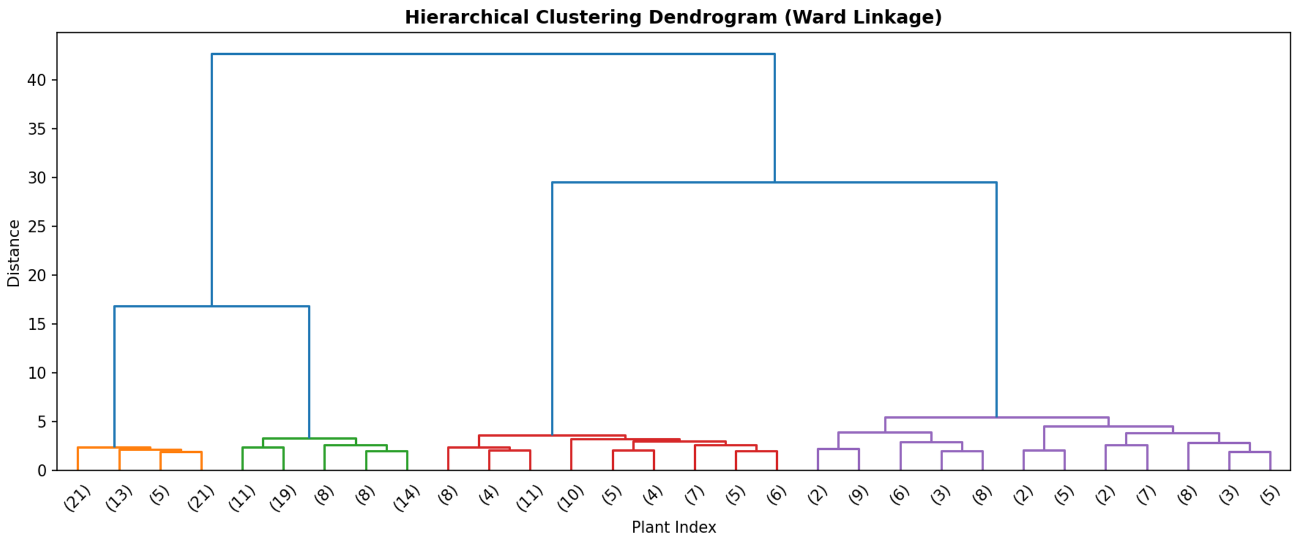
Cluster	Plant Type Match	Speed	Defect %	Energy	Labor	Downtime	Label / Interpretation
0	Labor-Intensive (n=60)	60.8	5.41	55.6	86.8	8.1	High labor cost, moderate defects, low energy — inefficient labor
1	Lean Manufacturing (n=60)	82.3	2.17	65.1	54.3	3.3	Best overall efficiency — high speed, low defects, minimal downtime
2	High Automation (n=60)	91.3	1.46	89.0	36.1	4.2	Highest speed and quality but high energy — automation tradeoff
3	Aging Facility (n=60)	57.5	7.65	92.7	69.9	14.3	Worst on all metrics — highest defects, energy, and downtime

Perfect 1:1 correspondence with known plant types (100%, 60 plants each). This is a remarkable result: the operational metrics alone fully recover the true plant categories without using any type label during clustering.

7. Figures







8. Conclusions

- **Research question answered — YES.** Four distinct, well-separated clusters were identified (Silhouette = 0.61), each perfectly matching a known plant type (100% accuracy), confirming both the 4-cluster solution and its external validity.
- **Aging Facilities (Cluster 3) are the highest-priority improvement target:** highest defect rate (7.65%), highest energy waste (92.7), highest downtime (14.3 hrs), yet lowest production speed (57.5). All metrics point to systemic failure.
- **Lean Manufacturing plants (Cluster 1) represent best practice:** high speed (82.3), low defects (2.17%), low labor cost (54.3), minimal downtime (3.3 hrs). These operational patterns should be studied and replicated.
- **High Automation plants (Cluster 2)** achieve the best speed (91.3) and lowest defects (1.46%) but consume the most energy (89.0). Energy optimization is their primary improvement lever.
- **Both K-Means and hierarchical clustering agree (Silhouette = 0.6096 for both), and PCA explains 89.5% of variance in 2D** — confirming the solution is robust, stable, and not an artifact of the clustering method.